AI-Assisted Kidney Stone Detection & Classification Report

Research Examination ID: VERIFY_2026_01 | Date: 2026-10-26 20:31:39

1. Image & Study Metadata

Parameter	Value
Examination ID	VERIFY_2026_01
Modality	CT Scan
Image Format	JPEG
DICOM Calibrated	No (Standard 2D Image)

2. Individual Deep Learning Classification Models

Model Architecture	Status	Predicted Class	Confidence	Stone Prob.	Non-Stone Prob.	Latency
MOBILENET V2	Trained & Loaded	**Non-Stone**	74.5%	25.5%	74.5%	29.73 ms
RESNET50	Trained & Loaded	**Non-Stone**	91.9%	8.1%	91.9%	45.4 ms
EFFICIENTNET B0	Trained & Loaded	**Non-Stone**	83.8%	16.2%	83.8%	33.5 ms

3. Multi-Model Ensemble Analysis

Ensemble Decision	**Non-Stone** (83.4% confidence)
Stone Probability	16.6%
Non-Stone Probability	83.4%
Contributing Classifiers	mobilenet_v2, resnet50, efficientnet_b0
Normalized Model Weights	{'mobilenet_v2': 0.3333, 'resnet50': 0.3333, 'efficientnet_b0': 0.3333}
Model Disagreement	Full Consensus

4. YOLOv8 Object Detection (Localization)

Detection model not trained or unavailable. Stone localization bounding-box annotations required.

5. Physical Measurements & Calibrations

No physical measurement calibrated for this examination.

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This report was generated by an academic research artificial intelligence system. The classification outputs, probability scores, and ensemble results are algorithmic estimates and do NOT constitute a medical diagnosis, clinical radiologist verification, or treatment recommendation. Model confidence must not be interpreted as an absolute probability of clinical disease. Clinical diagnosis requires thorough review by a qualified radiologist or physician.